

Doping Stability Report -- Ca in NaCoO₂ at 300 C

Generated: 2026-05-30 05:01 Host: NaCoO₂ Dopant: Ca

Executive Summary

Ca preferentially occupies the Na layer. The formation energy difference between the two layers is +0.220 eV, strongly favouring the Na site. MD confirms this preference: on the Na site the dopant MSD plateau slope is -0.00149 Å²/ps (stable), compared to 0.00162 Å²/ps on the Co site.

Site	E _f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Na	-2.102	+1	No	PASS	FAIL	STRUCTURAL COLLAPSE
Co	-1.882	-1	No	PASS	PASS	STABLE

Ca at the Na layer -- STRUCTURAL COLLAPSE

Charge situation

Ca replaces Na with a charge surplus of +1. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E _f	-2.1024 eV	PASS	< 1.0 eV to pass
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E_f = -2.102 eV (approximate -- charge surplus not modelled in CHGNet). This value is strongly negative, indicating that Ca incorporation is thermodynamically very favorable at the Na site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	-0.00149 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.041 Å²	----	
Max displacement	0.670 Å	----	
Coordination (min/mean/max)	3 / 5.7 / 7	FAIL	relaxed = 6, tolerance +/-1
Mean bond length (MD vs relaxed)	2.283 Å (relaxed: 2.2)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = -0.00149 Å²/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.041 Å²; maximum displacement from starting position = 0.67 Å.

Coordination fluctuated between 3 and 7 (mean 5.7), deviating by more than +/-1 from the relaxed value (6). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 2.283 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact

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rather than a true structural collapse. The Ca atom barely moves (MSD = 0.041 Å²) but the coordination count fluctuates because the bond length at the Na site is close to the analysis cutoff. Consider increasing coordination_cutoff_Å and re-running analysis.

? **WARNING:** Charge surplus (+1) cannot be modelled in CHGNet. Running single substitution without compensation. E_f is approximate.

Ca at the Co layer -- STABLE

Charge situation

Ca replaces Co with a charge mismatch of -1. To restore charge neutrality, 1 Na atom(s) were removed from the supercell (furthest from the dopant site). The formation energy accounts for the Na-vacancy term.

Formation Energy (thermodynamic stability)

Formation energy E _f	-1.8824 eV	PASS	< 1.0 eV to pass
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E_f = -1.882 eV. This value is strongly negative, indicating that Ca incorporation is thermodynamically very favorable at the Co site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	0.00162 Å²/ps	PASS	< 0.005 to pass (plateau = stable)
Final MSD	0.070 Å²	----	
Max displacement	0.696 Å	----	
Coordination (min/mean/max)	5 / 6.0 / 6	PASS	relaxed = 6, tolerance +/-1
Mean bond length (MD vs relaxed)	2.169 Å (relaxed: 2.1)	----	
Lattice volume change	+0.00 %	PASS	< +/-5% to pass

Late-time MSD slope = 0.00162 Å²/ps. The slope is small and below the migration threshold -- the dopant is site-stable. Final MSD = 0.070 Å²; maximum displacement from starting position = 0.70 Å.

Coordination fluctuated between 5 and 6 (mean 6.0), within the +/-1 tolerance of the relaxed value (6). This is consistent with normal thermal vibration. Mean nearest-neighbour distance during MD = 2.169 Å.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STABLE

Ca is thermodynamically favorable and kinetically stable on the Co site at 300 C. All four stability criteria pass: formation energy is favorable, the dopant does not migrate during MD, its coordination environment is preserved, and the host lattice volume is unchanged.

? **WARNING:** Charge deficit (-1) cannot be compensated in CHGNet (would require adding Na interstitials or oxidising Co). Running single substitution without compensation. E_f is approximate.

Site Preference Comparison

The Na site is preferred by 0.220 eV over the Co site. Formation energy: -2.102 eV (Na) vs -1.882 eV (Co).

Ca preferentially occupies the Na layer. The formation energy difference between the two layers is +0.220 eV, strongly favouring the Na site. MD confirms this preference: on the Na site the dopant MSD plateau slope is -0.00149 Å²/ps (stable), compared to 0.00162 Å²/ps on the Co site.

Caveats and Limitations

? CHGNet is charge-neutral.

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch $\neq 0$) are approximate even when structural compensation defects are included.

? No Freysoldt/Kumagai correction.

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? Metal-rich reference state.

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? MD timescales are short (25 ps default).

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? Energy rankings are more trustworthy than absolute values.

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.